

TEPS School Distinction - Analysis (Final Revision)

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```
## Knitr global settings
knitr::opts_chunk$set(echo = TRUE)
```

```
## Load packages
library(tidyverse)
library(haven)
library(fastDummies)
library(nnet)
library(stargazer)
library(modelsummary)
```

```
## Read data
teps <- read_dta("data/TEPS3.dta")
#glimpse(teps)
```

```
## Factorize and rename variables
teps$geoarea <- teps$geoarea %>% as.factor()
levels(teps$geoarea) <- c("Taipei", "Taoyuan", "Taichung", "Tainan", "Kaohsiung", "Taitung")
```

```
teps$schtype <- as.factor(teps$schtype)
levels(teps$schtype) <- c("ordi", "priv", "star")
```

```
teps$pses <- as.factor(teps$pses)
levels(teps$pses) <- c("E0I0", "E0I1", "E1I0", "E1I1")
teps$pses <- relevel(teps$pses, ref = "E0I1")
```

```
teps$pedu <- as.factor(teps$pedu)
teps$income <- as.factor(teps$income)
teps <- teps %>% rename(pedu_bi = pedu2, income_bi = income2)
```

```
teps$pethn <- as.factor(teps$pethn)
teps$pocc <- as.factor(teps$pocc)
```

```
## Check school type distribution
teps %>% group_by(sch_id) %>%
  summarise(sch_id = first(sch_id), schtype = first(schtype), geoarea = first(geoarea)) %>%
  xtabs(~ schtype + geoarea, .)
```

```
##          geoarea
```

```
## schtype Taipei Taoyuan Taichung Tainan Kaohsiung Taitung
##   ordi      78      24      52      23      29      18
##   priv      18       7       9      14       6       0
##   star      19       6      13       6       7       4
```

Score Distribution

```
teps %>% group_by(schtype) %>% summarise(
  n = n(),
  missing = sum(is.na(cogability)),
  cog_avg = mean(cogability, na.rm = TRUE),
  cog_std = sd(cogability, na.rm = TRUE),
  cog_Q1 = quantile(cogability, probs = c(0.25), na.rm = TRUE),
  cog_Q2 = median(cogability, na.rm = TRUE),
  cog_Q3 = quantile(cogability, probs = c(0.75), na.rm = TRUE)) %>%
  mutate_if(is.numeric, round, digits=3)
```

schtype	n	missing	cog_avg	cog_std	cog_Q1	cog_Q2	cog_Q3
ordi	13071	28	-0.201	0.995	-0.857	-0.112	0.503
priv	2306	4	0.498	0.860	-0.008	0.549	1.098
star	3676	5	0.397	0.917	-0.132	0.472	1.018

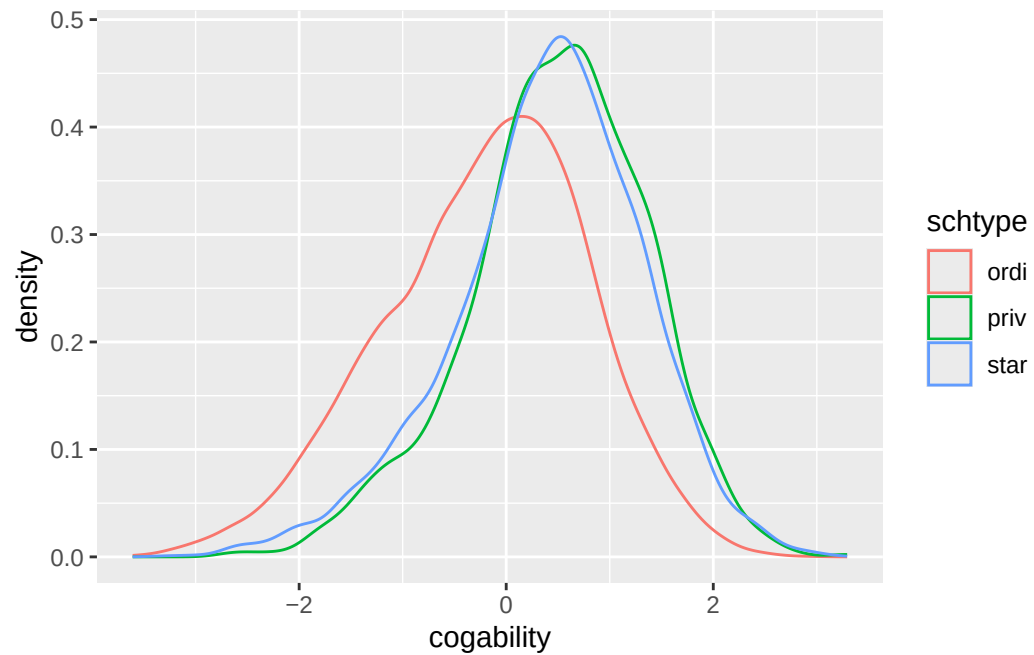
```
teps %>% group_by(schtype) %>% summarise(
  n = n(),
  missing = sum(is.na(finscore)),
  score_avg = mean(finscore, na.rm = TRUE),
  score_std = sd(finscore, na.rm = TRUE),
  score_Q1 = quantile(finscore, probs = c(0.25), na.rm = TRUE),
  score_Q2 = median(finscore, na.rm = TRUE),
  score_Q3 = quantile(finscore, probs = c(0.75), na.rm = TRUE)) %>%
  mutate_if(is.numeric, round, digits=1)
```

schtype	n	missing	score_avg	score_std	score_Q1	score_Q2	score_Q3
ordi	13071	975	144.8	70.9	82	141	205
priv	2306	558	193.7	59.0	153	204	240
star	3676	264	190.6	65.5	142	205	244

Visualization

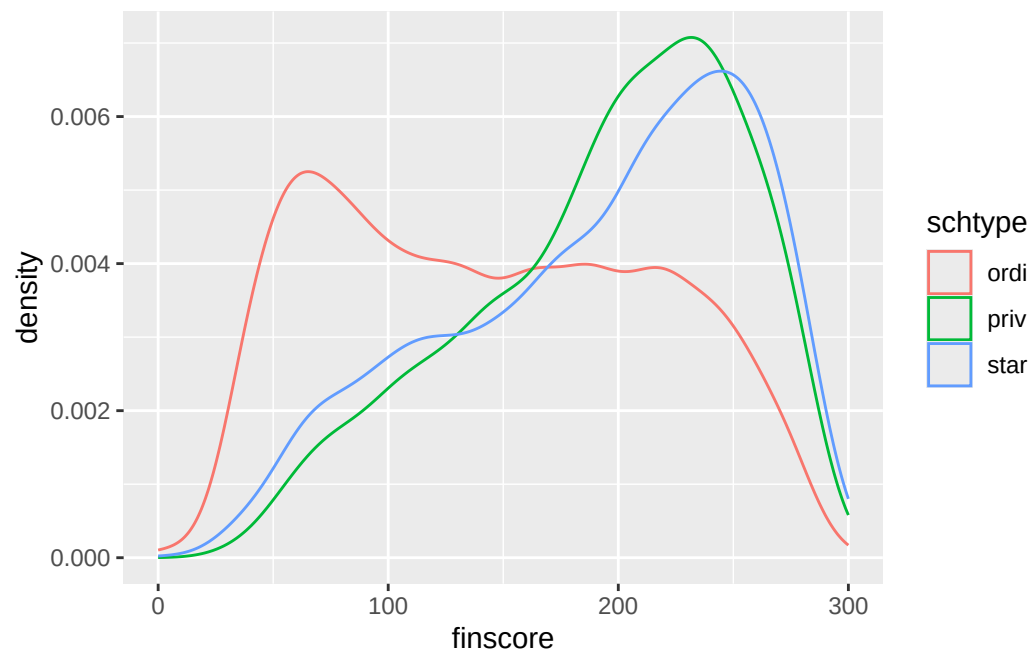
```
teps %>% ggplot(aes(x = cogability, color = schtype)) + geom_density()
```

```
## Warning: Removed 37 rows containing non-finite outside the scale range
## ('stat_density()').
```



```
teps %>% ggplot(aes(x = finscore, color = schtype)) + geom_density()
```

```
## Warning: Removed 1797 rows containing non-finite outside the scale range
## ('stat_density()').
```



```
#teps %>% filter(geoarea == "Taipei") %>% ggplot(aes(x = finscore, color = schtype)) + geom_density()
#teps %>% filter(geoarea == "Taichung") %>% ggplot(aes(x = finscore, color = schtype)) + geom_density()
```

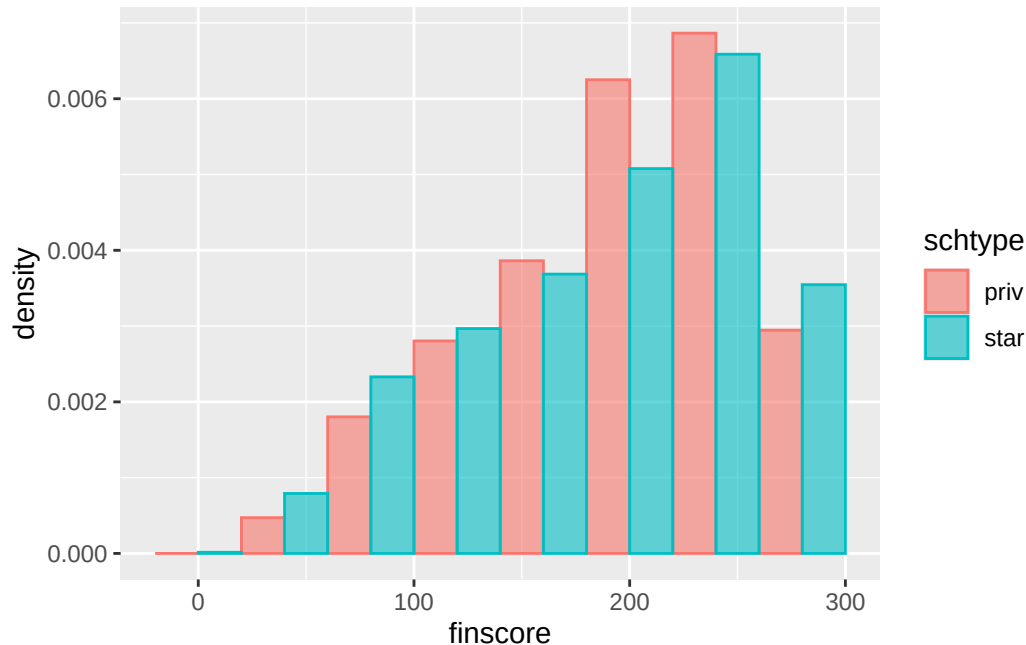
```

tpeps %>% filter(schtype != "ordi") %>%
  ggplot(aes(x = finscore, color = schtype, fill = schtype, after_stat(density))) +
  geom_histogram(position = "dodge", binwidth = 40, alpha = 0.6)

```

```
## Warning: Removed 822 rows containing non-finite outside the scale range
```

```
## ('stat_bin()').
```



```
## Select variables
```

```

tpeps <- tpeps %>%
  select(stud_id, sch_id, geoarea, priv, star, schtype,
         male, onlychild, edstchild, ygstchild, cogability, marriage, pethn,
         pedu, pedu_bi, income, income_bi, pses, pocc, pboss, pmgm,
         pexpedu, book, art, reloc,
         acadcomm, talentcls, abstudy, edusav)

```

```
## Drop NA
```

```

tpeps <- tpeps %>% drop_na() # n = 19053 -> 18465
#sapply(tpeps, function(y) sum(is.na(y))) %>% data.frame(na_count = .)

```

```
## Variable Distributions
```

```
tpeps %>% count(geoarea) %>% mutate(prop = n / sum(n))
```

geoarea	n	prop
Taipei	6358	0.3443271
Taoyuan	1767	0.0956946
Taichung	4752	0.2573517
Tainan	2183	0.1182237

geoarea	n	prop
Kaohsiung	2309	0.1250474
Taitung	1096	0.0593555

```
teps %>% count(pedu) %>% mutate(prop = n / sum(n))
```

pedu	n	prop
1	4953	0.2682372
2	7896	0.4276198
3	3111	0.1684809
4	1857	0.1005686
5	648	0.0350934

```
teps %>% count(income) %>% mutate(prop = n / sum(n))
```

income	n	prop
1	1824	0.0987815
2	7589	0.4109938
3	6500	0.3520173
4	1652	0.0894666
5	900	0.0487409

1. Description

```
## Make dummies for factors
teps2 <- teps %>% dummy_cols(select_columns = c("pethn", "pses", "pocc")) %>%
  select(-stud_id, -sch_id, -geoarea, -schtype, -pethn, -pedu, -income, -pses, -pocc)

## Calculate means
vars <- teps2 %>% colnames()
sum1 <- teps2 %>% summarise_all(mean)
sum2 <- teps2 %>% group_by(priv, star) %>% summarise_all(mean)

## Create a data frame
sumstat <- data.frame(
  Variable = vars,
  Mean_All = as.numeric(sum1),
  Ordinary = as.numeric(sum2[1,]),
  Private = as.numeric(sum2[3,]),
  Star = as.numeric(sum2[2,]))

# Output
sumstat <- sumstat %>% mutate_if(is.numeric, round, digits=3)
sumstat
```

Variable	Mean_All	Ordinary	Private	Star
priv	0.122	0.000	1.000	0.000
star	0.194	0.000	0.000	1.000
male	0.513	0.512	0.550	0.496
onlychild	0.058	0.051	0.083	0.066
edstchild	0.345	0.329	0.397	0.369
ygstchild	0.367	0.364	0.344	0.390
cogability	0.008	-0.191	0.501	0.400
marriage	0.906	0.892	0.944	0.928
pedu_bi	0.136	0.080	0.230	0.273
income_bi	0.138	0.083	0.294	0.234
pboss	0.145	0.110	0.295	0.175
pmgm	0.178	0.138	0.272	0.259
pexpedu	0.478	0.397	0.685	0.632
book	0.354	0.319	0.429	0.430
art	0.129	0.108	0.150	0.186
reloc	0.324	0.270	0.404	0.463
acadcomm	0.256	0.239	0.339	0.264
talentcls	0.657	0.591	0.830	0.781
abstudy	0.066	0.043	0.153	0.092
edusav	0.636	0.592	0.798	0.689
pethn_1	0.733	0.741	0.760	0.689
pethn_2	0.056	0.063	0.034	0.046
pethn_3	0.042	0.033	0.054	0.067
pethn_4	0.136	0.120	0.143	0.190
pethn_5	0.032	0.043	0.009	0.008
pses_E0I1	0.073	0.052	0.166	0.089
pses_E0I0	0.791	0.868	0.604	0.638
pses_E1I0	0.071	0.049	0.102	0.127
pses_E1I1	0.065	0.031	0.128	0.145
pocc_1	0.200	0.242	0.106	0.112
pocc_2	0.243	0.236	0.257	0.258
pocc_3	0.113	0.088	0.162	0.172
pocc_4	0.223	0.187	0.305	0.299
pocc_5	0.221	0.247	0.171	0.160

```
write_csv(sumstat, "export/summary_statistics.csv")
```

2. Private Schools

```
## M1: Independent
Priv1 <- teps %>%
  glm(priv ~ pedu_bi + income_bi,
       family = "binomial", data = .)

## M2: Independent + Control
Priv2 <- teps %>%
  glm(priv ~ pedu_bi + income_bi +
       male + onlychild + edstchild + ygstchild + cogability + marriage +
```

```

    pethn + geoarea + pocc + pboss + pmgm,
    family = "binomial", data = .)

## M3: Interaction
teps$peduXincome <- teps$pedu_bi*teps$income_bi
Priv3 <- teps %>%
  glm(priv ~ pedu_bi + income_bi + peduXincome +
       male + onlychild + edstchild + ygstchild + cogability + marriage +
       pethn + geoarea + pocc + pboss + pmgm,
       family = "binomial", data = .)

## M4: Categorization
Priv4 <- teps %>%
  glm(priv ~ pses +
       male + onlychild + edstchild + ygstchild + cogability + marriage +
       pethn + geoarea + pocc + pboss + pmgm,
       family = "binomial", data = .)

## Results
stargazer(Priv1, Priv2, Priv3, Priv4,
           star.cutoffs = c(0.05, 0.01, 0.001),
           order = c("pedu_bi", "income_bi", "peduXincome"),
           omit = c("geoarea"),
           type = "text", title = "Private vs. Public")

```

```

##
## Private vs. Public
## =====
##                               Dependent variable:
##                               -----
##                               priv
##                               (1)      (2)      (3)      (4)
## -----
## pedu_bi          0.313***    -0.083    0.207*
##                  (0.063)    (0.072)    (0.088)
##
## income_bi        1.026***    0.628***    0.857***
##                  (0.058)    (0.065)    (0.077)
##
## peduXincome              -0.671***
##                          (0.129)
##
## psesE0I0                      -0.857***
##                               (0.077)
##
## psesE1I0                      -0.650***
##                               (0.105)
##
## psesE1I1                      -0.463***
##                               (0.103)

```

##				
## male	0.204***	0.207***	0.207***	
##	(0.049)	(0.049)	(0.049)	
##				
## onlychild	0.644***	0.637***	0.637***	
##	(0.106)	(0.106)	(0.106)	
##				
## edstchild	0.203**	0.199**	0.199**	
##	(0.069)	(0.070)	(0.070)	
##				
## ygstchild	0.029	0.031	0.031	
##	(0.070)	(0.071)	(0.071)	
##				
## cogability	0.462***	0.465***	0.465***	
##	(0.029)	(0.029)	(0.029)	
##				
## marriage	0.353**	0.353**	0.353**	
##	(0.109)	(0.109)	(0.109)	
##				
## pethn2	-0.190	-0.188	-0.188	
##	(0.138)	(0.138)	(0.138)	
##				
## pethn3	-0.081	-0.058	-0.058	
##	(0.112)	(0.111)	(0.111)	
##				
## pethn4	-0.219**	-0.217**	-0.217**	
##	(0.071)	(0.071)	(0.071)	
##				
## pethn5	0.236	0.242	0.242	
##	(0.241)	(0.241)	(0.241)	
##				
## pocc2	0.530***	0.509***	0.509***	
##	(0.086)	(0.086)	(0.086)	
##				
## pocc3	0.832***	0.810***	0.810***	
##	(0.099)	(0.100)	(0.100)	
##				
## pocc4	0.649***	0.631***	0.631***	
##	(0.087)	(0.088)	(0.088)	
##				
## pocc5	0.418***	0.401***	0.401***	
##	(0.091)	(0.091)	(0.091)	
##				
## pboss	0.940***	0.904***	0.904***	
##	(0.059)	(0.060)	(0.060)	
##				
## pmgm	0.400***	0.395***	0.395***	
##	(0.060)	(0.060)	(0.060)	
##				
## Constant	-2.225***	-3.791***	-3.809***	-2.952***
##	(0.027)	(0.147)	(0.148)	(0.164)
##				
##	-----	-----	-----	-----
## Observations	18,465	18,465	18,465	18,465

```
## Log Likelihood      -6,627.651 -5,774.527 -5,761.105 -5,761.105
## Akaike Inf. Crit. 13,261.300 11,597.050 11,572.210 11,572.210
## =====
## Note:                *p<0.05; **p<0.01; ***p<0.001
```

```
modelsummary(list(Priv1, Priv2, Priv3, Priv4),
  stars = TRUE, output = "export/private.xlsx")
```

3. Star Schools

```
## Multinom
Star1 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  data = .)
```

```
## # weights: 78 (50 variable)
## initial value 20285.875910
## iter 10 value 14512.162506
## iter 20 value 13802.242081
## iter 30 value 13447.088303
## iter 40 value 13310.652599
## iter 50 value 13256.967502
## final value 13250.752291
## converged
```

```
## Multinom with a different reference group
teps$schtype <- relevel(teps$schtype, ref = "priv")
Star2 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  data = .)
```

```
## # weights: 78 (50 variable)
## initial value 20285.875910
## iter 10 value 14255.294740
## iter 20 value 13957.533234
## iter 30 value 13495.596695
## iter 40 value 13318.764061
## iter 50 value 13266.800868
## iter 60 value 13251.024942
## iter 70 value 13250.752567
## final value 13250.752350
## converged
```

```
## Results
stargazer(Star1, Star2,
  star.cutoffs = c(0.05, 0.01, 0.001),
```

```
omit = c("geoarea"),
type = "text", title = "Private & Star vs. Public")
```

```
##
## Private & Star vs. Public
## =====
##                               Dependent variable:
##                               -----
##                               priv      star      ordi      star
##                               (1)      (2)      (3)      (4)
##                               -----
## pedu_bi          0.502***    0.779***   -0.502***    0.277**
##                  (0.093)    (0.073)    (0.093)    (0.097)
##
## income_bi        1.030***    0.562***   -1.030***   -0.468***
##                  (0.082)    (0.078)    (0.082)    (0.091)
##
## peduXincome      -0.613***    -0.194     0.613***    0.418**
##                  (0.140)    (0.121)    (0.140)    (0.143)
##
## male             0.198***    -0.031    -0.198***   -0.229***
##                  (0.050)    (0.041)    (0.050)    (0.056)
##
## onlychild        0.727***    0.308**   -0.727***   -0.418***
##                  (0.111)    (0.096)    (0.111)    (0.123)
##
## edstchild        0.243***    0.172**   -0.243***    -0.071
##                  (0.072)    (0.058)    (0.072)    (0.081)
##
## ygstchild        0.097       0.239***   -0.097       0.142
##                  (0.072)    (0.057)    (0.072)    (0.081)
##
## cogability       0.609***    0.481***   -0.609***   -0.129***
##                  (0.030)    (0.024)    (0.030)    (0.033)
##
## marriage         0.378***    0.129     -0.378***   -0.248*
##                  (0.112)    (0.081)    (0.112)    (0.125)
##
## pethn2           -0.194      -0.014     0.194       0.180
##                  (0.141)    (0.100)    (0.141)    (0.157)
##
## pethn3           0.082       0.384***   -0.082       0.302*
##                  (0.118)    (0.093)    (0.118)    (0.122)
##
## pethn4           -0.127      0.259***    0.127       0.386***
##                  (0.074)    (0.056)    (0.074)    (0.079)
##
## pethn5           0.106      -1.240***   -0.106      -1.346***
##                  (0.246)    (0.201)    (0.246)    (0.307)
##
## pocc2            0.662***    0.684***   -0.662***    0.022
##                  (0.088)    (0.068)    (0.088)    (0.101)
##
```

```
## pocc3          0.987***   0.741***  -0.987***   -0.246*
##              (0.103)    (0.081)   (0.103)    (0.115)
##
## pocc4          0.790***   0.694***  -0.790***   -0.096
##              (0.090)    (0.071)   (0.090)    (0.103)
##
## pocc5          0.461***   0.316***  -0.461***   -0.145
##              (0.093)    (0.072)   (0.093)    (0.108)
##
## pboss          1.023***   0.386***  -1.023***  -0.636***
##              (0.063)    (0.058)   (0.063)    (0.070)
##
## pmgm           0.491***   0.300***  -0.491***   -0.191**
##              (0.063)    (0.053)   (0.063)    (0.068)
##
## Constant      -3.952***  -2.813***   3.951***   1.139***
##              (0.152)    (0.113)   (0.152)    (0.171)
##
## -----
## Akaike Inf. Crit. 26,601.500 26,601.500 26,601.500 26,601.500
## =====
## Note:                *p<0.05; **p<0.01; ***p<0.001
```

```
modelsummary(list(Star1, Star2),
              shape = term ~ model + response,
              stars = TRUE, output = "export/star.xlsx")
```

5. Mechanism

```
# Define dependent variables for the models (mediators)
dep_vars <- c("pexpedu", "book", "art", "reloc",
              "acadcomm", "talentcls", "abstudy", "edusav")

# Loop through each dependent variable and save models separately
for (i in seq_along(dep_vars)) {
  formula <- as.formula(paste(dep_vars[i], "~ pedu_bi + income_bi + peduXincome +
                              male + onlychild + edstchild + ygstchild +
                              marriage + pethn + geoarea +
                              pocc + pboss + pmgm"))
  model <- glm(formula, family = "binomial", data = teps)
  assign(paste0("Medi", i), model)
}

## Results
stargazer(Medi1, Medi2, Medi3, Medi4,
          dep.var.labels = dep_vars[1:4],
          star.cutoffs = c(0.05, 0.01, 0.001),
          keep = c("pedu_bi", "income_bi", "peduXincome"),
          type = "text", title = "Mediators")
```

```
##
## Mediators
## =====
##                               Dependent variable:
##                               -----
##                               pexpedu    book    art    reloc
##                               (1)      (2)    (3)    (4)
## -----
## pedu_bi      1.154***    0.475***    0.530***    0.230***
##              (0.073)    (0.062)    (0.078)    (0.063)
##
## income_bi    0.623***    0.222***    0.242**    0.384***
##              (0.064)    (0.062)    (0.085)    (0.062)
##
## peduXincome   0.015      -0.050    0.003      -0.300**
##              (0.133)    (0.101)    (0.125)    (0.101)
## -----
## Observations    18,465    18,465    18,465    18,465
## Log Likelihood  -11,280.490 -11,525.450 -6,776.969 -11,265.400
## Akaike Inf. Crit. 22,608.980 23,098.900 13,601.940 22,578.790
## =====
## Note:                                *p<0.05; **p<0.01; ***p<0.001
```

```
stargazer(Medi5, Medi6, Medi7, Medi8,
  dep.var.labels = dep_vars[5:8],
  star.cutoffs = c(0.05, 0.01, 0.001),
  keep = c("pedu_bi", "income_bi", "peduXincome"),
  type = "text", title = "Mediators")
```

```
##
## Mediators
## =====
##                               Dependent variable:
##                               -----
##                               acadcomm    talentcls    abstudy    edusav
##                               (1)      (2)    (3)    (4)
## -----
## pedu_bi      0.067      0.887***    0.638***    0.152*
##              (0.068)    (0.085)    (0.104)    (0.069)
##
## income_bi    0.299***    0.637***    0.952***    0.841***
##              (0.065)    (0.076)    (0.093)    (0.075)
##
## peduXincome  -0.355**    -0.045    -0.372**    -0.315*
##              (0.109)    (0.157)    (0.144)    (0.125)
## -----
## Observations    18,465    18,465    18,465    18,465
## Log Likelihood  -10,338.690 -10,743.230 -3,952.167 -11,556.380
## Akaike Inf. Crit. 20,725.380 21,534.460 7,952.334 23,160.760
## =====
## Note:                                *p<0.05; **p<0.01; ***p<0.001
```

```
## Multinom
teps$schtype <- relevel(teps$schtype, ref = "ordi")
Mech1 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
            pexpedu + book + art + reloc +
            acadcomm + talentcls + abstudy + edusav +
            male + onlychild + edstchild + ygstchild + cogability + marriage +
            pethn + geoarea + pocc + pboss + pmgm,
            data = .)
```

```
## # weights: 102 (66 variable)
## initial value 20285.875910
## iter 10 value 14766.011921
## iter 20 value 13987.348502
## iter 30 value 13387.023626
## iter 40 value 12996.715683
## iter 50 value 12945.508139
## iter 60 value 12932.044806
## iter 70 value 12927.507102
## iter 80 value 12920.908794
## final value 12920.833224
## converged
```

```
## Multinom with a different reference group
teps$schtype <- relevel(teps$schtype, ref = "priv")
Mech2 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
            pexpedu + book + art + reloc +
            acadcomm + talentcls + abstudy + edusav +
            male + onlychild + edstchild + ygstchild + cogability + marriage +
            pethn + geoarea + pocc + pboss + pmgm,
            data = .)
```

```
## # weights: 102 (66 variable)
## initial value 20285.875910
## iter 10 value 13805.942289
## iter 20 value 13592.829764
## iter 30 value 13209.872151
## iter 40 value 13074.876275
## iter 50 value 12943.846748
## iter 60 value 12924.182974
## iter 70 value 12921.049590
## iter 80 value 12920.842637
## final value 12920.833267
## converged
```

```
## Results
stargazer(Mech1, Mech2,
           star.cutoffs = c(0.05, 0.01, 0.001),
           omit = c("geoarea", "pethn", "pocc"),
           type = "text", title = "Mechanism")
```

```
##
```

## Mechanism				
## =====				
##	Dependent variable:			
##	-----			
##	priv	star	ordi	star
##	(1)	(2)	(3)	(4)
##	-----			
## pedu_bi	0.331***	0.657***	-0.331***	0.327***
##	(0.095)	(0.075)	(0.095)	(0.098)
##				
## income_bi	0.797***	0.443***	-0.797***	-0.355***
##	(0.084)	(0.079)	(0.084)	(0.093)
##				
## peduXincome	-0.490***	-0.121	0.490***	0.369*
##	(0.142)	(0.122)	(0.142)	(0.145)
##				
## pexpedu	0.463***	0.300***	-0.463***	-0.162*
##	(0.058)	(0.046)	(0.058)	(0.065)
##				
## book	-0.012	0.054	0.012	0.065
##	(0.055)	(0.045)	(0.055)	(0.060)
##				
## art	-0.153*	0.186**	0.153*	0.340***
##	(0.076)	(0.059)	(0.076)	(0.080)
##				
## reloc	0.130*	0.557***	-0.130*	0.428***
##	(0.054)	(0.043)	(0.054)	(0.058)
##				
## acadcomm	0.272***	-0.065	-0.272***	-0.337***
##	(0.056)	(0.048)	(0.056)	(0.063)
##				
## talentcls	0.543***	0.318***	-0.543***	-0.226**
##	(0.067)	(0.050)	(0.067)	(0.076)
##				
## abstudy	0.662***	0.152	-0.662***	-0.510***
##	(0.086)	(0.081)	(0.086)	(0.090)
##				
## edusav	0.533***	0.026	-0.533***	-0.507***
##	(0.061)	(0.045)	(0.061)	(0.068)
##				
## male	0.219***	0.008	-0.219***	-0.211***
##	(0.052)	(0.041)	(0.052)	(0.057)
##				
## onlychild	0.604***	0.184	-0.604***	-0.420***
##	(0.113)	(0.097)	(0.113)	(0.125)
##				
## edstchild	0.180*	0.100	-0.180*	-0.080
##	(0.073)	(0.059)	(0.073)	(0.082)
##				
## ygstchild	0.089	0.226***	-0.089	0.137
##	(0.074)	(0.058)	(0.074)	(0.082)
##				
## cogability	0.507***	0.395***	-0.507***	-0.112**
##	(0.032)	(0.025)	(0.032)	(0.035)

```
##
## marriage          0.352**    0.103    -0.352**    -0.249*
##                   (0.114)    (0.082)    (0.114)    (0.127)
##
## pboss             0.869***    0.286***   -0.869***   -0.583***
##                   (0.065)    (0.059)    (0.065)    (0.072)
##
## pmgm              0.362***    0.218***   -0.362***   -0.143*
##                   (0.064)    (0.054)    (0.064)    (0.069)
##
## Constant          -4.851***   -3.239***    4.851***    1.612***
##                   (0.165)    (0.120)    (0.165)    (0.185)
##
## -----
## Akaike Inf. Crit. 25,973.670 25,973.670 25,973.670 25,973.670
## =====
## Note:                                *p<0.05; **p<0.01; ***p<0.001
```

```
modelsummary(list(Mech1, Mech2),
  shape = term ~ model + response,
  stars = TRUE, output = "export/mechanism.xlsx")
```

```
## Multinom
teps$schtype <- relevel(teps$schtype, ref = "ordi")
Mech3 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    pexpedu + book + art + reloc +
    acadcomm + talentcls + abstudy + edusav +
    male + onlychild + edstchild + ygstchild + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  data = .)
```

```
## # weights: 99 (64 variable)
## initial value 20285.875910
## iter 10 value 15550.952896
## iter 20 value 14634.512787
## iter 30 value 13599.470481
## iter 40 value 13200.084104
## iter 50 value 13148.175542
## iter 60 value 13142.795115
## iter 70 value 13138.144186
## iter 80 value 13134.341736
## final value 13134.331843
## converged
```

```
## Multinom with a different reference group
teps$schtype <- relevel(teps$schtype, ref = "priv")
Mech4 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    pexpedu + book + art + reloc +
    acadcomm + talentcls + abstudy + edusav +
    male + onlychild + edstchild + ygstchild + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  data = .)
```

```
## # weights: 99 (64 variable)
## initial value 20285.875910
## iter 10 value 13796.292774
## iter 20 value 13575.954250
## iter 30 value 13365.360650
## iter 40 value 13248.941009
## iter 50 value 13154.729851
## iter 60 value 13141.566526
## iter 70 value 13136.477930
## iter 80 value 13134.339043
## final value 13134.331869
## converged
```

Results

```
stargazer(Mech3, Mech4,
  star.cutoffs = c(0.05, 0.01, 0.001),
  keep = c("pedu_bi", "income_bi", "peduXincome",
    "pexpedu", "book", "art", "reloc",
    "acadcomm", "talentcls", "abstudy", "edusav"),
  type = "text", title = "Mechanism w/o Cogability")
```

```
##
## Mechanism w/o Cogability
## =====
##                               Dependent variable:
##                               -----
##                               priv      star      ordi      star
##                               (1)       (2)       (3)       (4)
## -----
## pedu_bi          0.427***    0.735***   -0.427***    0.308**
##                  (0.094)     (0.074)     (0.094)     (0.098)
##
## income_bi        0.807***    0.449***   -0.807***   -0.359***
##                  (0.083)     (0.078)     (0.083)     (0.093)
##
## peduXincome      -0.413**     -0.057     0.413**     0.356*
##                  (0.141)     (0.121)     (0.141)     (0.144)
##
## pexpedu          0.689***    0.478***   -0.689***   -0.212***
##                  (0.056)     (0.044)     (0.056)     (0.064)
##
## book             0.040       0.100*     -0.040       0.060
##                  (0.054)     (0.044)     (0.054)     (0.060)
##
## art             -0.150*     0.184**     0.150*     0.335***
##                  (0.076)     (0.059)     (0.076)     (0.080)
##
## reloc            0.154**     0.583***   -0.154**     0.430***
##                  (0.053)     (0.042)     (0.053)     (0.058)
##
## acadcomm         0.223***    -0.107*    -0.223***   -0.330***
##                  (0.056)     (0.047)     (0.056)     (0.062)
##
## talentcls        0.648***    0.400***   -0.648***   -0.248***
```

```

##          (0.066)    (0.049)    (0.066)    (0.075)
##
## abstudy    0.629***    0.116    -0.629***    -0.513***
##          (0.084)    (0.080)    (0.084)    (0.090)
##
## edusav     0.541***    0.032    -0.541***    -0.509***
##          (0.061)    (0.044)    (0.061)    (0.068)
##
## -----
## Akaike Inf. Crit. 26,396.660 26,396.660 26,396.660 26,396.660
## =====
## Note:                *p<0.05; **p<0.01; ***p<0.001

```
